Alexander J. Elliott

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Video Editing • Robotics • Guitar • Photography • Physical AI • Climbing • Geography

EXPERIENCE

Magna International • Mechatronics CO-OP *Jan - Aug 2025 • Newmarket, ON*

- Developed Python and TypeScript tools to automate pre- and post-processing for FEA, multibody dynamics, and inertial analysis, improving simulation efficiency
- Architected and deployed Inertia+, automating inertia calculations and engineering report generation, reducing engineering effort by up to 26%
- Developed an engineering visualization platform enabling collaborative review of simulation results and faster design iteration

Key Skills: CAE • FEA Workflows • Simulation Automation • Python • TypeScript • Engineering Analysis

AME Group • Mechanical CO-OP & QAQC CO-OP *Sept - Dec 2023, May - Aug 2024 • Vancouver, BC*

- Automated drafting and engineering workflows using Python, PyAutoCAD, and Revit plugins
- Developed C# automation tools to generate schematics, convert AutoCAD details, and improve productivity
- Led BIM improvement initiatives that standardized engineering workflows and enabled teams to work more efficiently within Revit

Key Skills: Mechanical Design • Revit • AutoCAD • BIM • Engineering Automation • Process Improvement

UBC Uncrewed Aircraft Systems • Mechanical Payload Team *Sep 2022 – Sep 2024 • Vancouver, BC*

- Designed payload deployment systems using electromagnetic clutches, viscous damping, and guided parachutes
- Designed and manufactured carbon fibre payload structures for the aircraft
- Contributed to a team placing 2nd and 3rd in the 2024 AEAC competition

Key Skills: Mechanical Design • Composite Manufacturing • UAV Systems • Payload Integration • Rapid Prototyping

PROJECTS

Litter Collecting Drone • UBC Mech 458 Capstone *Sep 2025 – Apr 2026*

- Designed and manufactured a custom UAV platform for autonomous litter retrieval in challenging environments
- Designed the architecture of a quadcopter incorporating outward-canted motors to reduce downdraft during retrieval
- Manufactured and assembled the complete aircraft platform

Key Skills: Robotics • Computer Vision • YOLOv8 • UAV • Embedded AI • Systems Integration

Automated Spice Dispensing Robot • UBC Mech 423 Final Project *Jan – Apr 2026*

- Designed the mechanical carousel-style architecture for a robotic dispensing system, inspired by CNC tool changers
- Developed embedded motor-control firmware on a custom MSP430-based PCB to drive a stepper motor and enable closed-loop DC motor positioning
- Integrated sensing, actuation, and desktop software, including IR sensor and limit switch feedback
- Created a voice-enabled C# WinForms interface for fast, reliable controls, with a focus on user convenience

Key Skills: Embedded Systems • Robotics • Motor Control • Sensors • C • C#

EDUCATION

University of British Columbia *2021 - 2026 • Vancouver, BC*

BASc Mechanical Engineering • Mechatronics Option • w/ Co-Op • w/ Distinction • 5x Dean's List

SKILLS

Mechanical Engineering: Mechanical Design • DFM • Rapid Prototyping • Engineering Simulation • Vehicle Dynamics
CAD & Engineering Software: SolidWorks • CATIA • Fusion 360 • Onshape • Revit • AutoCAD • MATLAB
Manufacturing: CNC Machining • 3D Printing • Composite Manufacturing • Soldering • EtherCAT • Mach4
Embedded & Controls: Arduino • MSP430 • Raspberry Pi • Motor Control • Sensors • PCB Design